

Exercise 2

SQL Aggregate functions and operators

1. SELECT DISTINCT department
FROM student;

department
IT
HR
Finance

2. SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;

department	avg_age
IT	20.5
HR	22
Finance	23

3. SELECT departments, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

department	student_count
IT	2
HR	2

4. SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student_id, name, age, department
FROM students
WHERE (department = 'IT' OR department = 'HR')
AND age > 21;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
5	Eve	22	HR

6. SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

department	total_credits
IT	8

7. SELECT course_id, course_name, department, credits
FROM courses
WHERE credits NOT < 4;

course_id	course_name	department	credits
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	credits
101	SQL Basics	4
102	Python	4
103	Data Science	4

9. SELECT MAX(grade) AS max_grade
MIN(grade) AS min_grade
AVG(grade) AS avg_grade
FROM enrollments;

max_grade	min_grade	avg_grade
90	78	84.8

10. SELECT course_id, COUNT(*) AS enrollment-
count
FROM enrollments
GROUP BY course_id;

course id	enrollment count
101	1
102	1
103	1
104	1
105	1

11. SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus

FROM Salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	170 000	6 000

12. SELECT department, AVG(salary) AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG(salary) > 55 000